

Human Photo Classifier

Detect and sort photos of people

User Manual

Automatically find photos that contain people using YOLO, and sort them by face and gender.

Designed by Codecider Lab

Open-source • Windows • GPU accelerated

1. Overview

Human Photo Classifier is a desktop application that scans a folder of images, detects which photos contain people, and copies them into a destination folder. Optionally, it can further sort person photos by **face presence** and **gender** into subfolders.

Key features

- Person detection powered by YOLO (Ultralytics).
- Optional face detection and gender recognition (InsightFace).
- Automatic sorting into category subfolders.
- Corrupt images are excluded; partially damaged photos are set aside.
- GPU acceleration with automatic CPU fallback.
- Responsive dark-themed interface; the app never freezes during processing.

2. Installation & First Run

Requirements

- Windows 10/11, Python 3.10 or newer.
- An NVIDIA GPU is optional but recommended for speed.

Setup

Double-click **run.bat**. On the first run it automatically creates a virtual environment, installs the dependencies, and launches the app. Later runs start the app directly.

On first use, the detection models are downloaded automatically: the YOLO weights (`yolo11n.pt`) and the InsightFace model pack (`buffalo_l`). An internet connection is required the first time.

Running the tests (optional)

Double-click **test.bat** to run the unit test suite.

3. The Interface

The window has two columns. The left column holds the settings and controls; the right column shows the live log.

Area	Purpose
Folders	Choose the Source folder (input) and Destination folder (output).
Detection	Model, Confidence, Image size, and Device settings.
Classification	Optional Face detection and Gender recognition.
Start / Stop	Begin or safely halt processing.
Progress bar	Shows how many images have been processed.

Log	Per-image results and the final summary.
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4. Step-by-Step Usage

- 1 Click **Browse** next to **Source** and select the folder of images to scan.
- 2 Click **Browse** next to **Destination** and select where results should be saved (must differ from Source).
- 3 Adjust the **Detection** settings if needed (defaults work well).
- 4 Optionally enable **Face detection** and/or **Gender recognition**.
- 5 Click **Start**. Watch progress in the log; click **Stop** to halt at any time.
- 6 When finished, the summary line reports totals, and results are in the Destination folder.

5. Detection Settings

Setting	Meaning	Default
Model	YOLO model. Larger = more accurate but slower (n < s < m < l).	yolo11n.pt
Confidence	Minimum score to count as a person. Lower = catches more, more false positives.	0.20
Image size	Inference resolution. Larger detects small/distant people better, but is slower.	960
Device	cpu, or cuda / cuda:0 for GPU acceleration.	cpu
Search subfolders	Also scan nested folders and mirror their structure in the output.	On

Tip: To catch as many people as possible, lower Confidence to 0.10–0.20, raise Image size to 1280+, and use a larger model such as yolo11s.pt or yolo11m.pt.

6. Classification Options

These options sort person photos into subfolders inside the Destination folder. Enabling **Gender recognition** automatically enables **Face detection**, because gender is inferred from a detected face. When several faces are present, the largest (main subject) face decides the gender.

Face detection	Gender recognition	Subfolders created
Off	Off	None (person photos extracted, source structure kept)
On	Off	with_face / no_face / unclassified / others
Off	On	male / female / unclassified / others
On	On	with_face/male, with_face/female, no_face, unclassified, others

7. Output Folder Structure

Category folders (in English) are created under your Destination folder:

Folder	Contents
with_face	Person photos where a face was detected.
no_face	Person photos with no detectable face (e.g. back view).
male / female	Person photos sorted by the main subject's gender.
unclassified	Person photos whose face analysis failed.
others	Photos that do not contain a person.
others/damaged	Person or other photos that are partially damaged.

Example, with both options enabled:

```
Destination/  
  with_face/  
    male/    2024/party.jpg  
    female/  portrait.jpg  
  no_face/   back_view.jpg  
  others/    landscape.jpg  
  others/damaged/ truncated.jpg
```

8. Damaged & Corrupt Images

Every image is checked before processing:

Condition	Handling
OK	Header and pixels decode fully → classified normally.
Partially damaged	Opens but pixel data is truncated/broken → copied to others/damaged.
Fully corrupt	Cannot be opened at all → excluded (not copied).

Damaged photos are separated so they never mix into the clean category folders. In extraction-only mode (classification off), damaged images are skipped.

9. GPU Acceleration

- **Person detection (YOLO):** choose `cuda` or `cuda:0` in the Device setting to use the GPU.
- **Face & gender (InsightFace):** uses the GPU automatically when `onnxruntime-gpu` is installed, otherwise falls back to CPU (logged).
- The required CUDA 12 / cuDNN 9 runtime libraries are reused from the bundled PyTorch (cu124) install, so no separate CUDA Toolkit is needed.

10. Troubleshooting

Symptom	What to do
"Python not found"	Install Python 3.10+ and add it to PATH, then rerun run.bat.
No images found	Check the Source folder and enable Search subfolders.
Slow processing	Set Device to cuda, or use a smaller model / image size.
Few people detected	Lower Confidence and raise Image size; use a larger model.
Face model runs on CPU	Install onnxruntime-gpu for GPU face/gender analysis.

11. Command-Line Interface

The same pipeline is available without the GUI, for scripting and batch jobs. On Windows you can use `cli.bat` in place of `python cli.py`.

```
python cli.py SOURCE DESTINATION [options]
```

```
python cli.py ./photos ./sorted
```

```
python cli.py ./photos ./sorted --device cuda --face --gender
```

```
python cli.py ./photos ./sorted --model yolo11s.pt --confidence 0.15 -q
```

Option	Description
--model	YOLO model file (default yolo11n.pt).
--confidence	Detection confidence 0–1 (default 0.20).
--image-size	Inference image size (default 960).
--device	cpu, cuda, or cuda:0 (default cpu).
--no-recursive	Do not scan subfolders.
--face	Sort person photos into with_face / no_face.
--gender	Sort person photos by gender (male / female).
-q, --quiet	Show a progress bar instead of per-image lines.

Press **Ctrl+C** to stop gracefully; a summary is still printed. The exit code is 0 on success and 1 on a fatal error.

Human Photo Classifier is open-source software, released under the GNU AGPL-3.0. It uses Ultralytics YOLO (AGPL-3.0) and PyQt5 (GPL-3.0); the InsightFace models are for non-commercial research use only. Model weights are downloaded on first run. Designed by Codecider Lab.